

Practice Quiz 01D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 01

Constructing a formula from prescribed truth values.

1. Construct a formula using only the letters P, Q and the connectives $\neg$, $\wedge$, and $\vee$ that is true exactly when P and Q have the same truth value. Verify your formula with a complete four-row truth table.

Solution One possible formula is

$$T = (P \wedge Q) \vee (\neg P \wedge \neg Q).$$

Its first conjunction handles the case when both letters are true; its second handles the case when both are false.

P	Q	$P \wedge Q$	$\neg P \wedge \neg Q$	T
F	F	F	T	T
F	T	F	F	F
T	F	F	F	F
T	T	T	F	T

The final column is true exactly in the two rows in which P and Q agree, as required.

